# SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS MSC IN CONTROL & OPTIMISATION

1. Exercise

a) The vector field $f(x_1,x_2)$ is polynomial with respect to both variables. Hence it is a locally Lipschitz function and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [ 2 ]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathcal{N}_1 = \{x : x_2 = x_1(x_1^2-1)(x_1^2-4)\}.$$

Notice that $x_2$ can be expressed as a polynomial function of degree 5 in the variable $x_1$. In particular, the nullcline crosses the $x_1$ axis for $x_1 = 0, \pm 1, \pm 2$. For the second nullcline we have:

$$\mathcal{N}_2 = \{x : x_2 = -2x_1\}.$$

This is a straight line with negative slope $(-2)$ passing through the origin. [ 2 ]

c) The nullclines and the regions that they delimit are shown in Fig. 1.1. [ 4 ]

d) There are no forward invariant regions formed by the nullclines. Indeed each region has outgoing arrows in some parts of its boundary. [ 2 ]

e) To find equilibria we need to solve the system of equations $\dot{x}_1 = 0$, $\dot{x}_2 = 0$ in the unknowns $x_1$ and $x_2$. From the second equation we have $x_2 = -2_x 1$. Hence, substituting in the first equation we get:

$$-x_1(x_1^2-1)(x_1^2-4) - 2x_1 = -x_1^5 + 5x_1^3 - 6x_1 = 0$$

This admits one trivial root in $x_1 = 0$, and 4 additional roots obtained by solving:

$$-x_1^4 + 5x_1^2 - 6 = 0$$

Using $x_1^2$ as the unknown we get $x_1^2 = 2$ and $x_1^2 = 3$. In particular, then $x_1 = \pm\sqrt{2}$ and $x_1 = \pm\sqrt{3}$. There are 5 equilibria in total, in $(0,0)$, $\pm(\sqrt{2}, -2\sqrt{2})$, $\pm(\sqrt{3}, -2\sqrt{3})$. [ 3 ]

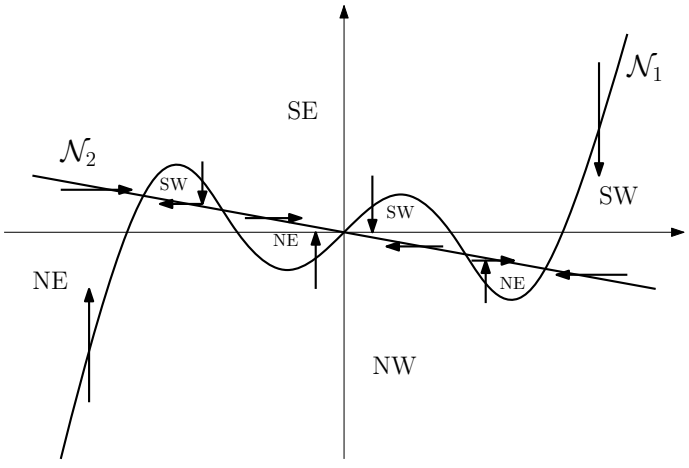



Figure 1.1 Regions formed by the nullclines and directions of the vector-field

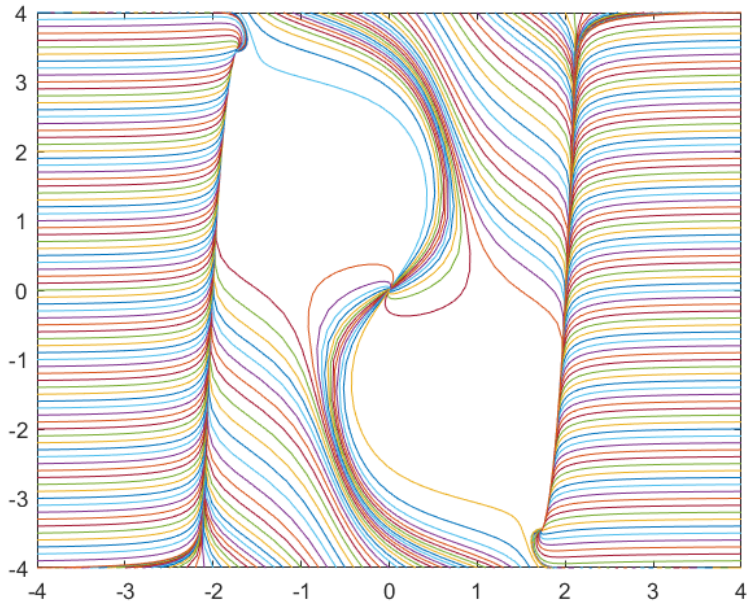



Figure 1.2 Global phase portrait

f) Computing the Jacobian of the vector-field $f$ yields:

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} -5x_1^4 + 15x_1^2 - 4 & 1 \\ -2 & -1 \end{bmatrix}.$$

Evaluating this in the origin we have:

$$\frac{\partial f}{\partial x}(0) = \begin{bmatrix} -4 & 1 \\ -2 & -1 \end{bmatrix},$$

whose characteristic polynomial is: $\chi(s) = (s+4)(s+1)+2 = s^2+5s+6$. Its roots are in $-3$ and $-2$, thus corresponding to a stable node (hyperbolic).
For the equilibria in $\pm(\sqrt{2}, -2\sqrt{2})$ we have:

$$\frac{\partial f}{\partial x}(\pm(\sqrt{2}, -2\sqrt{2})) = \begin{bmatrix} 6 & 1 \\ -2 & -1 \end{bmatrix},$$

whose characteristic polynomial is: $\chi(s) = (s-6)(s+1)+2 = s^2-5s-4$. Its roots are real and opposite in sign, $(5 \pm \sqrt{41})/2$, corresponding to the phase-portrait of a saddle. Finally, for equilibria in $\pm(\sqrt{3}, -2\sqrt{3})$ we have the same Jacobian of the equilibrium in 0, corresponding to a stable node. [ 4 ]

g) The phase portrait of the system is shown in Fig. 1.2. [ 3 ]

2. Exercise

a) We choose $x(t) = [z(t), \dot{z}(t)]^T = [x_1(t), x_2(t)]^T$. Accordingly we get the following state-space equations:

$$\dot{x}_1(t) = \dot{z}(t) = x_2(t)$$

$$\begin{aligned}\dot{x}_2(t) = \ddot{z}(t) &= -\dot{z}(t)z^2(t) - \dot{z}^3(t) - \sin(z(t)) + u(t) \\ &= -x_1^2(t)x_2(t) - x_2^3(t) - \sin(x_1(t)) + u(t)\end{aligned}$$

$$y = x_2(t)$$

[ 4 ]

b) We can write the system's dynamics in the following form:

$$\dot{x} = f(x) + g(x)u$$

with:

$$f(x) = \begin{bmatrix} x_2 \\ -x_1^2 x_2 - x_2^3 - \sin(x_1) \end{bmatrix}, \qquad g(x) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

which shows that this is an input-affine system. [ 2 ]

c) From the system's dynamics we see that:

$$\begin{aligned}yu = x_2[\dot{x}_2 + x_1^2 x_2 + x_2^3 + \sin(x_1)] &= x_2\dot{x}_2 + (x_1x_2)^2 + x_2^4 + \sin(x_1)\dot{x}_1 \\ &\geq \frac{d}{dt}\left[\frac{x_2^2}{2} - \cos(x_1)\right]\end{aligned}$$

Hence:

$$S(x) = \frac{x_2^2}{2} - \cos(x_1)$$

can be taken as a suitable storage function. Notice that $S(x) \geq -1$, viz. $S(\cdot)$ is bounded from below as required by storage functions. [ 4 ]

d) For $u = 0$ equilibria are given by the solutions of the equation $f(x) = 0$. These are all states fulfilling:

$$x_2 = 0, \qquad -x_1^2 x_2 - x_2^3 - \sin(x_1) = 0.$$

Solving the above equations yield: $x = [0, k\pi]$ for $k$ any integer. Consider the situation arising for $k = 1$. The Jacobian of the system is:

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -2x_1x_2 - \cos(x_1) & -x_1^2 - 3x_2^2 \end{bmatrix}$$

Evaluated in $[\pi, 0]$ yields:

$$\left.\frac{\partial f}{\partial x}\right|_{x=[\pi,0]} = \begin{bmatrix} 0 & 1 \\ 1 & -\pi^2 \end{bmatrix}.$$

The characteristic polynomial of this matrix is:

$$\chi(s) = s^2 + \pi^2 s - 1.$$

This admits two real roots of opposite sign. Hence it has an eigenvalue with positive real part and by Lyapunov's linearisation method the equilibrium of the original nonlinear system is unstable (viz. not Lyapunov stable). [ 4 ]

e) The feedback $u = -z$ transforms the $\dot{x}_2$ equation as follows:

$$\dot{x}_2 = -x_1^2 x_2 - x_2^3 - \sin(x_1) - x_1.$$

Consider the Lyapunov function:

$$V(x) = 1 + \frac{x_1^2}{2} - \cos(x_1) + \frac{x_2^2}{2}.$$

It is easy to see that

$$V(x) \geq \frac{x_1^2 + x_2^2}{2},$$

which proves that $V(x)$ is positive definite and radially unbounded. Moreover, for the closed-loop system:

$$\dot{V}(x) = [x_1 + \sin(x_1)]x_2 + x_2[-x_1^2 x_2 - x_2^3 - \sin(x_1) - x_1] = -x_1^2 x_2^2 - x_2^4 \leq 0.$$

Since $\dot{V}$ is negative semi-definite we compute its Kernel, as follows:

$$\text{Ker}[\dot{V}] = \{x : x_2 = 0\}.$$

Any invariant set included in the Kernel must also be included in:

$$\{x : x_2 = 0 \text{ and} \dot{x}_2 = 0\} = \{x : x_2 = 0 \text{ and} - x_1^2 x_2 - x_2^3 - \sin(x_1) - x_1 = 0\}$$

$$= \{x : x_2 = 0 \text{ and} - \sin(x_1) - x_1 = 0\} = \{0\}.$$

Hence, Lasalle's invariance principle applies in its global form and the origin is proven to be a globally asymptotically stable equilibrium. [ 6 ]

3. Exercise

a) Let $\Sigma_1$ be the scalar system of equations:

$$\dot{x}_1 = -x_1^3 + d_1$$

with state $x_1$ and input $d_1$. Similarly let $\Sigma_2$ be the scalar system of equations:

$$\dot{x}_2 = -x_2^3 + ad_2^9$$

with state $x_2$ and input $d_2$. The original system is recovered through the following feedback interconnection $d_1 = x_2$ and $x_2 = d_1$. [ 4 ]

b) Consider $\Sigma_1$. Let $V_1(x_1) = x_1^2/2$. Taking derivatives along solutions yields:

$$\dot{V}_1 = x_1(-x_1^3 + d1) \leq -x_1^4 + |x_1||d_1|.$$

In particular, for any $\varepsilon \in (0,1)$ we see that

$$(1-\varepsilon)|x_1|^4 \geq |x_1||d_1| \Rightarrow \dot{V}_1 \leq -\varepsilon|x_1|^4.$$

In particular then:

$$|x_1| \geq \sqrt[3]{|d_1|/(1-\varepsilon)} \Rightarrow \dot{V}_1 \leq -\varepsilon|x_1|^4.$$

This implies ISS with gain $\gamma_1(r) = \sqrt[3]{|r|/(1-\varepsilon)}$. [ 3 ]
Similarly for the $\Sigma_2$ subsystem we let $V_2(x_2) = x_2^2/2$. Taking derivatives yields:

$$\dot{V}_2 = x_2(-x_2^3 + ad_2^9) \leq -|x_2|^4 + |a||x_2||d_2|^9.$$

Notice that, for any $\varepsilon \in (0,1)$ we see that

$$(1-\varepsilon)|x_2|^4 \geq |a||x_2||d_2|^9 \Rightarrow \dot{V}_2 \leq -\varepsilon|x_2|^4.$$

In particular then:

$$|x_2| \geq \sqrt[3]{|a|/(1-\varepsilon)}|d_2|^3 \Rightarrow \dot{V}_2 \leq -\varepsilon|x_2|^4.$$

This implies ISS with gain $\gamma_2(r) = \sqrt[3]{|a|/(1-\varepsilon)}|d_2|^3$. [ 3 ]

c) The cascade of ISS systems is always ISS. To compute the gain from input to output notice that:

$$\limsup_{t\to+\infty} |x_2(t)| \leq \gamma_2(\limsup_{t\to+\infty} |d_2(t)|)$$

$$= \gamma_2(\limsup_{t\to+\infty} |x_1(t)|) \leq \limsup_{t\to+\infty} \gamma_2(\gamma_1(|d_1(t)|)).$$

Hence the gain of the series can be estimated to be: $\sqrt[3]{|a|/(1-\varepsilon)}|\gamma_1(r)|^3 = \sqrt[3]{|a|/(1-\varepsilon)}r/(1-\varepsilon)$. [ 3 ]

d) The gain of the series is also the loop gain. Notice that for all $|a| < 1$ there exists a sufficiently small $\varepsilon$ such that this is less than unity, viz.

$$\gamma_2(\gamma_1(r)) < r \qquad \forall r \neq 0.$$

Hence the system is GAS at the origin by virtue of the small gain theorem. [ 3 ]

e) From the asymptotic gain property we know that:

$$\limsup_{t\to+\infty} |x_1(t)| \leq \gamma_1(\limsup_{t\to+\infty} |d_1(t)|) = \gamma_1(1) = \sqrt[3]{1/(1-\varepsilon)}.$$

[ 4 ]

4. Exercise

a) To compute the relative degree we differentiate the output $y$ along solutions of the system:

$$\dot{y} = \dot{x}_1 + \dot{x}_2 = 2x_3$$

$$\ddot{y} = 2\dot{x}_3 = 2(1+x_1^2)^{-1}u + 2\cos(x_2)$$

Since the coefficient of $u$ is non-zero in $\ddot{y}$ throughout state-space, the relative degree is 2 and is globally defined. [ 2 ]

b) Letting $u = (1+x_1^2)[v/2 - \cos(x_2)]$ yields the closed-loop Input-Output dynamics:

$$\ddot{y} = v,$$

thus achieving Input-Output linearisation of the system. [ 3 ]

c) To stabilize a double-integrator it is enough to use a PD controller $v = -k_p y - k_d \dot{y}$ for any $k_p$ and $k_d$ positive. [ 3 ]

d) We choose the following state variables:

$$\xi_1 = x_1 + x_2, \qquad \xi_2 = 2x_3, \qquad \zeta = x_1 - x_2.$$

Accordingly the closed-loop equations read:

$$\begin{aligned} \dot{\xi}_1 &= \xi_2 \\ \dot{\xi}_2 &= v = -k_p\xi_1 - k_d\xi_2 \\ \dot{\zeta} &= -2\zeta^3 + 2y \\ \\ y &= \xi_1 \end{aligned}$$

These equations are in normal form. The $\dot{\zeta}$ equation describe the so called internal dynamics. [ 4 ]

e) If we regard $y$ as an exogenous disturbance in the $\zeta$ equation (viz. $d := y$), we are left with the following scalar nonlinear non-autonomous system:

$$\dot{\zeta} = -2\zeta^3 + 2d$$

To analyse its ISS, we choose $V(\zeta) := \zeta^2/2$. Accordingly the derivative reads:

$$\dot{V} = \zeta[-2\zeta^3 + 2d] \leq -|\zeta|^4 - |\zeta|^4 + 2|\zeta||d|.$$

The following implication holds:

$$|\zeta| \geq \sqrt[3]{2|d|} \Rightarrow \dot{V} \leq -|\zeta|^4$$

which proves that $V$ is an ISS Lyapunov function of type 2 for the internal dynamics. Hence the internal dynamics are ISS. [ 4 ]

f) The overall system can be seen as the cascade of the Input-Output dynamics and the internal dynamics. Since ISS is preserved under cascades (and the I-O dynamics are also ISS, being linear and GAS), the overall system is globally asymptotically stable (and in fact also ISS with respect to additive disturbances in the I-O dynamics). [ 4 ]